```
clc;
clear;
close all;

%% ============================================================
% FIR FILTER REALIZATION AND COMPARATIVE ANALYSIS
% Direct Realization and Cascade Realization
%% ============================================================

%% 1. FIR FILTER PARAMETERS

b = [0.0625 0.2500 0.3750 0.2500 0.0625];

Fs = 1000;                      % Sampling frequency (Hz)
N = 128;                        % Number of samples

x = [1 zeros(1, N-1)];         % Unit impulse input
n = 0:N-1;                      % Sample indices

%% 2. DIRECT REALIZATION

% Obtain the impulse response using direct FIR filtering
yd = filter(b, 1, x);

%% 3. CASCADE REALIZATION

% Find the zeros of the FIR filter
z = roots(b);

% Gain factor
g = b(1);

% Initialize cascade output
yc = g * x;

% Cascade of first-order sections
for k = 1:length(z)
    yc = filter([1 -z(k)], 1, yc);
end

% Remove negligible imaginary numerical components
yc = real(yc);

%% 4. SYMMETRY AND LINEAR-PHASE ANALYSIS

% Check coefficient symmetry
symErr = max(abs(b - fliplr(b)));

if symErr < 1e-12
```

```
    symmetry = 'Symmetric';
else
    symmetry = 'Not Symmetric';
end

% FIR filter order
FilterOrder = length(b) - 1;

% Group delay for a symmetric FIR filter
D = FilterOrder / 2;

%% 5. FREQUENCY, PHASE AND GROUP DELAY

[H, w] = freqz(b, 1, 1024);

% Unwrapped phase response
phase = unwrap(angle(H));

% Group delay in samples
gd = grpdelay(b, 1, 1024);

%% 6. NUMERICAL COMPARISON

% Maximum absolute error between realizations
err = max(abs(yd - yc));

% Reference response magnitude
reference = max(abs(yd));

% Percentage deviation
if reference > 0
    dev = (err / reference) * 100;
else
    dev = 0;
end

%% 7. DISPLAY RESULTS

fprintf('\n================================================\n');
```

```
================================================
```

```
fprintf('       FIR FILTER REALIZATION RESULTS\n');
```

```
       FIR FILTER REALIZATION RESULTS
```

```
fprintf('================================================\n');
```

```
================================================
```

```
fprintf('\nFIR Filter Coefficients:\n');
```

```
FIR Filter Coefficients:
```

```
disp(b);
```

```
    0.0625    0.2500    0.3750    0.2500    0.0625
```

```
fprintf('Sampling Frequency       = %d Hz\n', Fs);
```

```
Sampling Frequency       = 1000 Hz
```

```
fprintf('Number of Samples        = %d\n', N);
```

```
Number of Samples        = 128
```

```
fprintf('Filter Order             = %d\n', FilterOrder);
```

```
Filter Order             = 4
```

```
fprintf('Coefficient Symmetry     = %s\n', symmetry);
```

```
Coefficient Symmetry     = Symmetric
```

```
fprintf('Symmetry Error           = %.3e\n', symErr);
```

```
Symmetry Error           = 0.000e+00
```

```
fprintf('Theoretical Group Delay   = %.2f samples\n', D);
```

```
Theoretical Group Delay   = 2.00 samples
```

```
fprintf('Maximum Realization Error = %.3e\n', err);
```

```
Maximum Realization Error = 8.327e-16
```

```
fprintf('Percentage Deviation     = %.3e %%\n', dev);
```

```
Percentage Deviation     = 2.220e-13 %
```

```
%% 8. DISPLAY FIRST 10 IMPULSE RESPONSE VALUES

fprintf('\n================================================\n');
```

```
================================================
```

```
fprintf('       FIRST 10 IMPULSE RESPONSE VALUES\n');
```

```
       FIRST 10 IMPULSE RESPONSE VALUES
```

```
fprintf('================================================\n');
```

```
================================================
```

```
fprintf('%5s %15s %15s\n', 'n', 'Direct', 'Cascade');
```

```
    n          Direct         Cascade
```

```
for k = 1:10
    fprintf('%5d %15.6f %15.6f\n', ...
        n(k), yd(k), yc(k));
end
```

```
    0        0.062500        0.062500
    1        0.250000        0.250000
    2        0.375000        0.375000
    3        0.250000        0.250000
    4        0.062500        0.062500
    5        0.000000        0.000000
    6        0.000000        0.000000
    7        0.000000        0.000000
    8        0.000000        0.000000
    9        0.000000        0.000000
```

```
%% 9. DISPLAY CASCADE ZEROS

fprintf('\n================================================\n');
```

```
================================================
```

```
fprintf('           CASCADE REALIZATION ZEROS\n');
```

```
           CASCADE REALIZATION ZEROS
```

```
fprintf('================================================\n');
```

```
================================================
```

```
disp(z);
```

```
  -1.0002 + 0.0000i
  -1.0000 + 0.0002i
  -1.0000 - 0.0002i
  -0.9998 + 0.0000i
```

```
%% 10. CREATE OUTPUT PARAMETER TABLE

Parameter = {
    'Direct realization output'
    'Cascade realization output'
    'Impulse response'
    'Phase response'
    'Comparison of realizations'
    };

Observation = {
    'Direct FIR filtering'
    'Cascade of first-order sections'
    'Equal to FIR coefficient sequence'
    'Linear phase'
```

```
    'Equivalent transfer-function behavior'
    };

Value = {
    sprintf('[%.4f %.4f %.4f %.4f %.4f]', yd(1:5))
    sprintf('[%.4f %.4f %.4f %.4f %.4f]', yc(1:5))
    sprintf('[%.4f %.4f %.4f %.4f %.4f]', yd(1:5))
    sprintf('Group delay = %.2f samples', D)
    sprintf('Max error = %.3e; Deviation = %.3e %%', err, dev)
    };

Unit_Note = {
    'Amplitude'
    'Amplitude'
    'h[n]'
    'Samples'
    'Numerical precision'
    };

Output_Table = table(Parameter, Observation, Value, Unit_Note);

fprintf('\n================================================\n');
```

```
================================================
```

```
fprintf('            OUTPUT PARAMETER TABLE\n');
```

```
            OUTPUT PARAMETER TABLE
```

```
fprintf('================================================\n');
```

```
================================================
```

```
disp(Output_Table);
```

Parameter	Observation	V
{'Direct realization output' }	{'Direct FIR filtering' }	{'[0.0625 0.2500 0.3750
{'Cascade realization output'}	{'Cascade of first-order sections' }	{'[0.0625 0.2500 0.3750
{'Impulse response' }	{'Equal to FIR coefficient sequence' }	{'[0.0625 0.2500 0.3750
{'Phase response' }	{'Linear phase' }	{'Group delay = 2.00 sa
{'Comparison of realizations'}	{'Equivalent transfer-function behavior'}	{'Max error = 8.327e-16

```
%% 11. GRAPH 1: DIRECT VS CASCADE IMPULSE RESPONSE

figure;

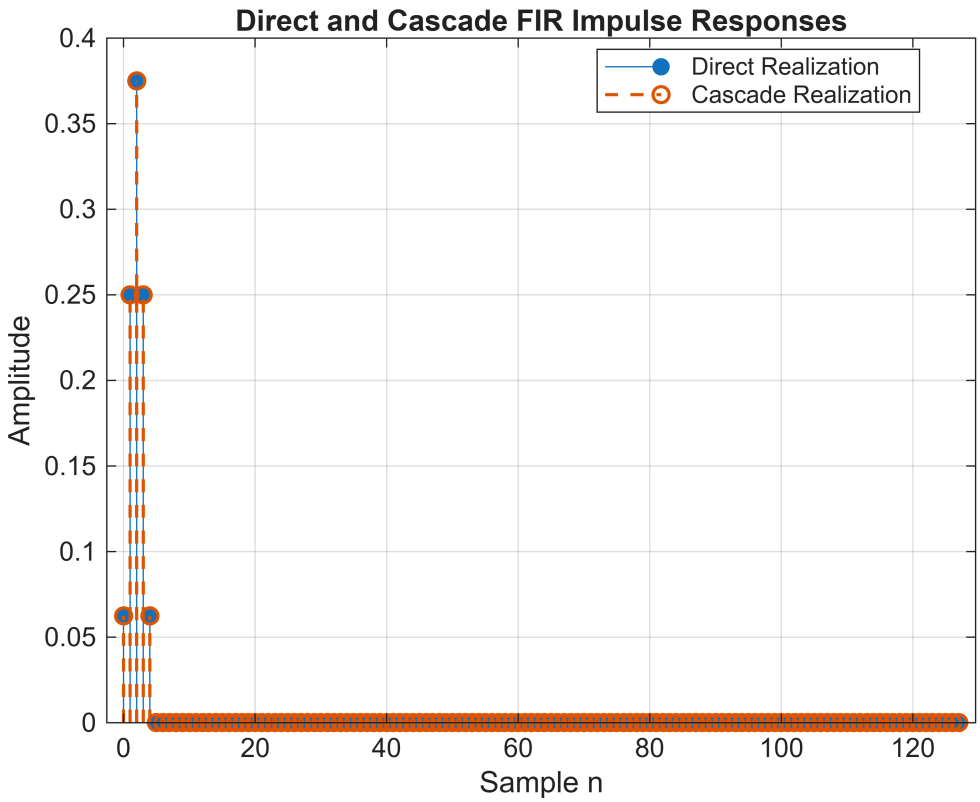
stem(n, yd, 'filled');
hold on;
stem(n, yc, '--', 'LineWidth', 1.2);
```

```
grid on;
xlabel('Sample n');
ylabel('Amplitude');

title('Direct and Cascade FIR Impulse Responses');

legend('Direct Realization', ...
       'Cascade Realization', ...
       'Location', 'best');

hold off;
```



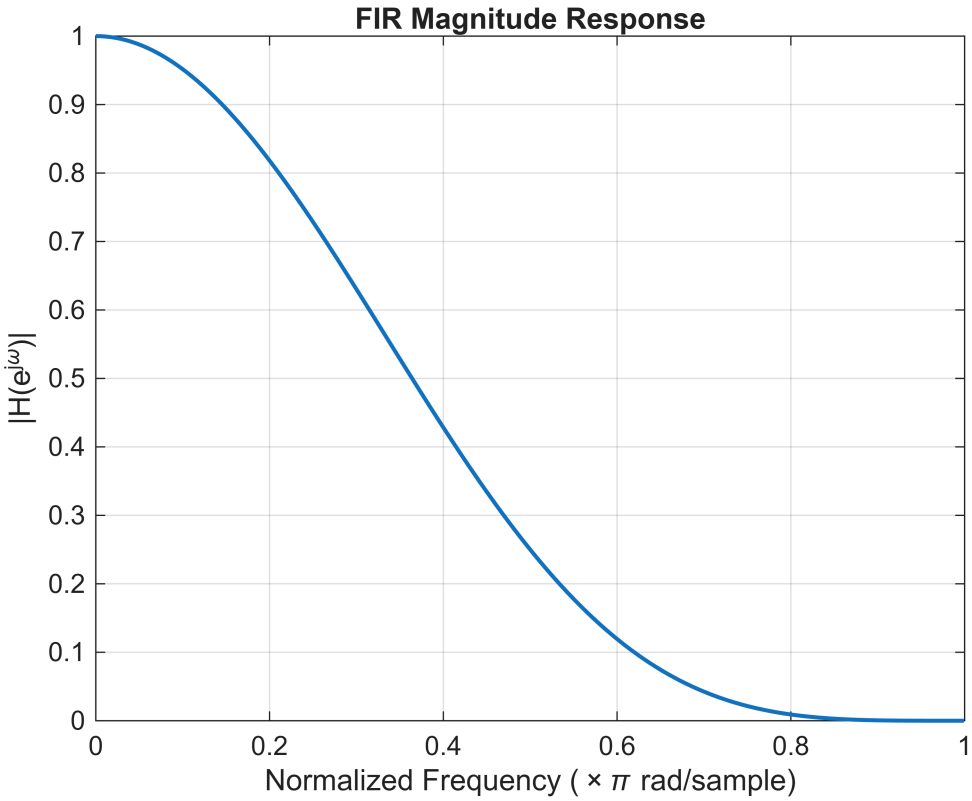

```
%% 12. GRAPH 2: MAGNITUDE RESPONSE

figure;

plot(w/pi, abs(H), 'LineWidth', 1.5);

grid on;
xlabel('Normalized Frequency (\times\pi rad/sample)');
ylabel('|H(e^{j\omega})|');

title('FIR Magnitude Response');
```



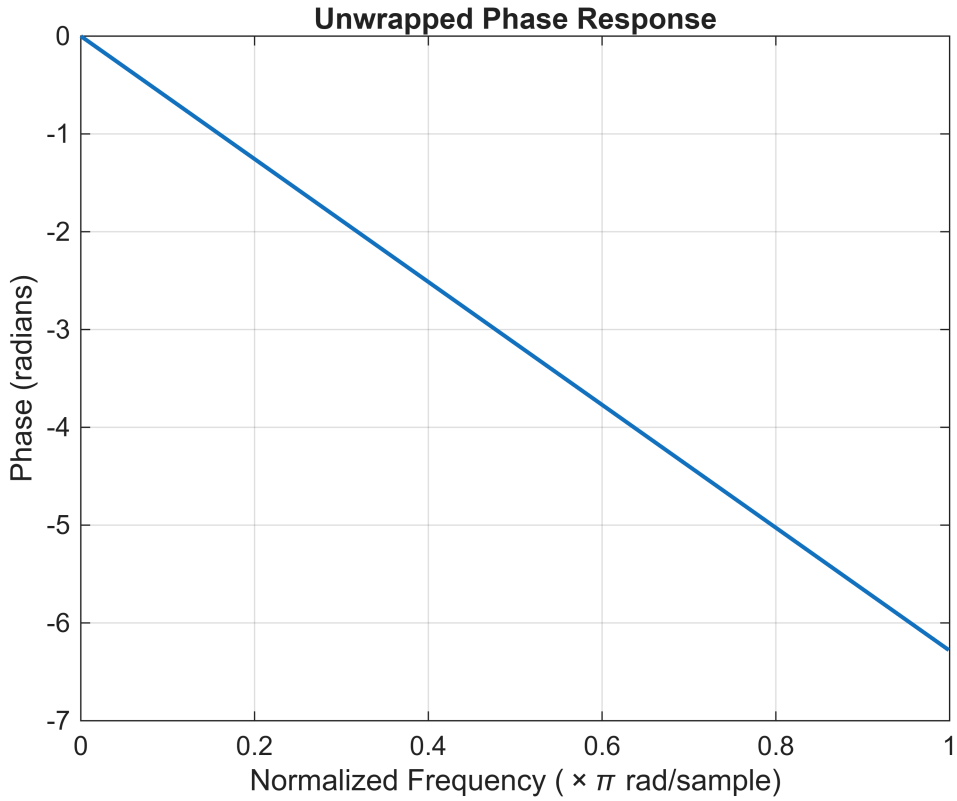

```
%% 13. GRAPH 3: PHASE RESPONSE

figure;

plot(w/pi, phase, 'LineWidth', 1.5);

grid on;
xlabel('Normalized Frequency (\times\pi rad/sample)');
ylabel('Phase (radians)');

title('Unwrapped Phase Response');
```



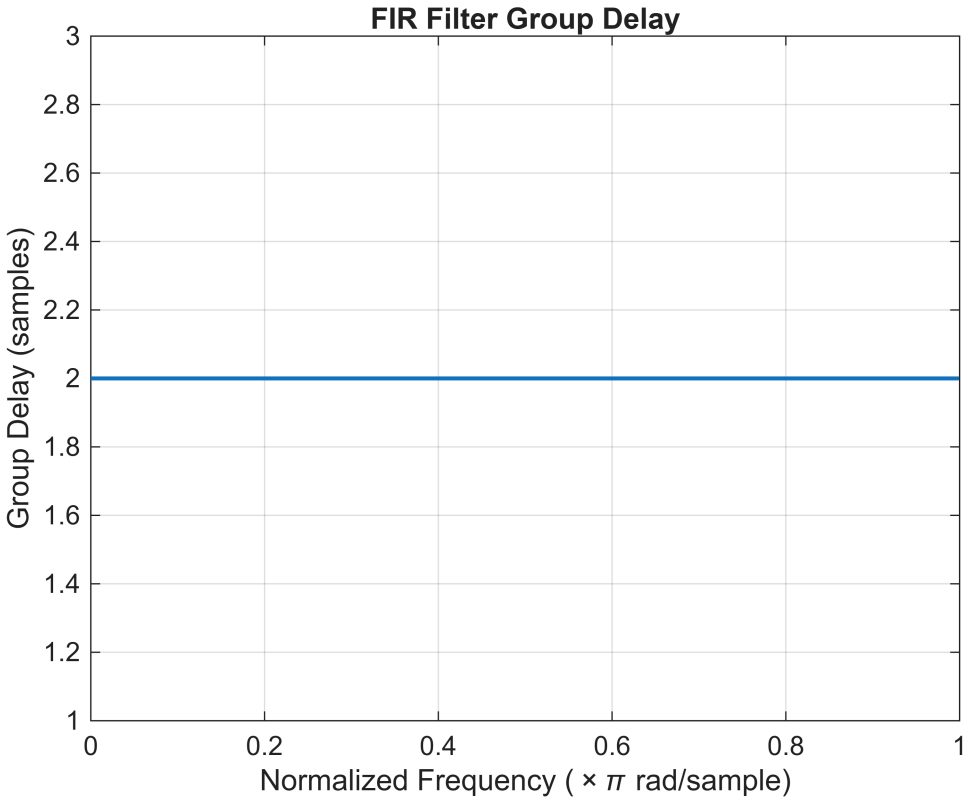

```
%% 14. GRAPH 4: GROUP DELAY

figure;

plot(w/pi, gd, 'LineWidth', 1.5);

grid on;
xlabel('Normalized Frequency (\times\pi rad/sample)');
ylabel('Group Delay (samples)');

title('FIR Filter Group Delay');
```



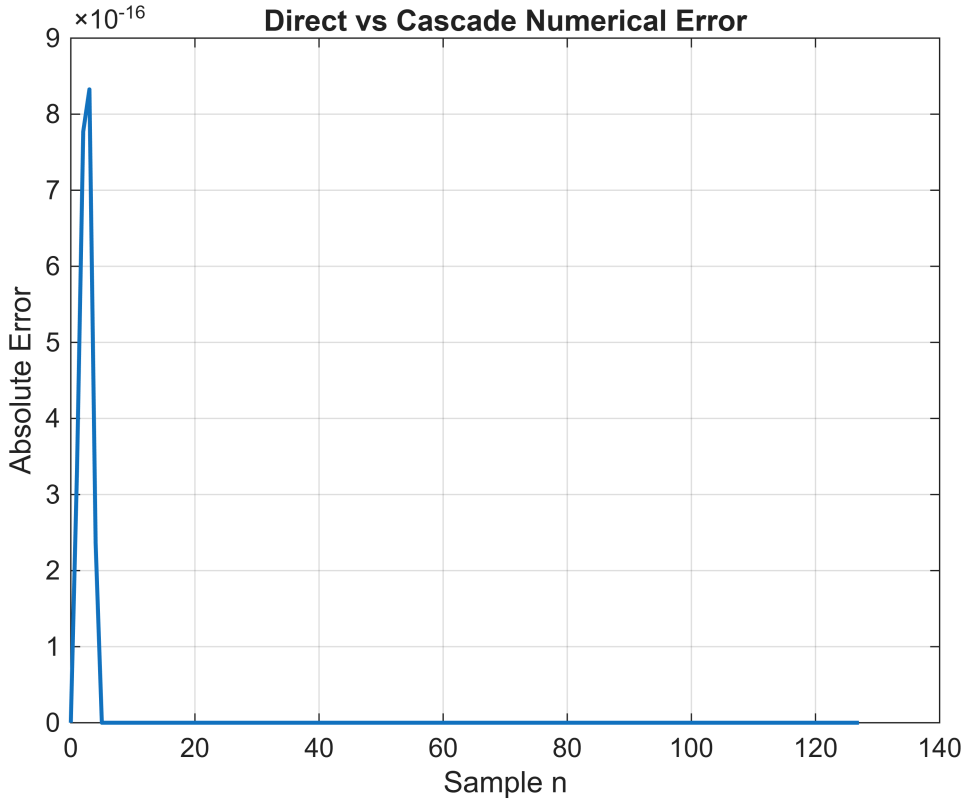

```
%% 15. GRAPH 5: NUMERICAL ERROR

figure;

plot(n, abs(yd - yc), 'LineWidth', 1.5);

grid on;
xlabel('Sample n');
ylabel('Absolute Error');

title('Direct vs Cascade Numerical Error');
```




```
%% 16. FINAL RESULT

fprintf('\n================================================\n');
```

```
================================================
```

```
fprintf('                  FINAL RESULT\n');
```

```
                  FINAL RESULT
```

```
fprintf('================================================\n');
```

```
================================================
```

```
fprintf('1. FIR filter realized using Direct structure.\n');
```

```
1. FIR filter realized using Direct structure.
```

```
fprintf('2. FIR filter realized using Cascade structure.\n');
```

```
2. FIR filter realized using Cascade structure.
```

```
fprintf('3. Coefficients are %s.\n', symmetry);
```

```
3. Coefficients are Symmetric.
```

```
fprintf('4. Linear-phase group delay = %.2f samples.\n', D);
```

```
4. Linear-phase group delay = 2.00 samples.
```

```
fprintf('5. Maximum realization error = %.3e.\n', err);
```

```
5. Maximum realization error = 8.327e-16.
```

```
fprintf('6. Percentage deviation = %.3e %%.\n', dev);
```

```
6. Percentage deviation = 2.220e-13 %.
```

```
fprintf('7. Direct and Cascade responses are equivalent.\n');
```

```
7. Direct and Cascade responses are equivalent.
```

```
fprintf('\n================================================\n');
```

```
================================================
```

```
fprintf('              END OF EXPERIMENT\n');
```

```
              END OF EXPERIMENT
```

```
fprintf('================================================\n');
```

```
================================================
```

```
```